

Notes: Brandt, Van Biesebroeck, Wang, and Zhang (JDE, 2026)

Where Has All the Dynamism Gone? Productivity Growth in China's Manufacturing Sector, 1998–2013

Contents

1	Big picture	3
2	Data and measurement (key objects)	3
2.1	NBS annual survey of above-scale industrial enterprises	3
2.2	STA annual tax survey	4
2.3	Matching the two sources and observed discrepancies	4
2.4	Additional data, variable construction, and applications	5
3	Models and empirical specifications (all equations + interpretation)	5
3.1	Target moment: aggregate productivity as a weighted expectation	5
3.2	Density-ratio reweighting	6
3.3	Constant versus time-varying weighting	6
3.4	Estimating the density ratio	6
3.5	Simulating representative samples	7
3.6	Production function estimation	7
3.7	Aggregation and productivity growth	8
3.8	Decomposition into old firms and young firms	8
4	Identification and instruments (what makes the estimates credible)	8
4.1	Why the standard data approach fails	8
4.2	What identifies the representative reweighting strategy	9
4.3	Why the validation exercises are persuasive	9
4.4	Why the production-function stage is credible	9
4.5	Main threats and limitations	9

4.6	Relation to the literature	10
5	Tables and figures	10
5.1	Table 1: Coverage of the NBS and STA samples	10
5.2	Table 2 and Figure 1: Reporting discrepancies across the two surveys	10
5.3	Figure 2 and Figure 3: Recovering the STA sampling scheme	12
5.4	Table 3 and Figure 4: How the simulated samples change the picture	13
5.5	Figure 5: Production elasticities before and after 2007	14
5.6	Figure 6: Aggregate TFP growth collapses after 2007	14
5.7	Figure 7 and Figure 8: Ownership and industry heterogeneity	14
5.8	Figure 9, Figure 10, and Figure 11: Regional patterns and convergence	15
5.9	Figure 12, Figure 13, Table 4, and Table 5: The weakening role of entrants	16
5.10	Table 6 and robustness checks	17
6	Short summary	18
7	Conclusion	18
7.1	Core conclusion (what the paper establishes)	18
7.2	Economic intuition	18
7.3	Takeaway	18

1 Big picture

- **Question.** How much did productivity growth in China’s manufacturing sector slow after 2007, and how can we measure that slowdown when the standard National Bureau of Statistics (NBS) firm survey becomes unreliable?
- **Setting.** The paper studies Chinese manufacturing over 1998–2013 by combining the traditional NBS above-scale industrial firm survey with an alternative annual tax survey collected by the State Taxation Administration (STA).
- **Core challenge.** The post-2007 NBS data have two major problems: important variables are missing and key reported quantities appear increasingly overstated. The STA tax data are cleaner for many purposes, but they are *not representative* and their sampling weights are unknown.
- **Main methodological result.** The paper develops a way to recover the implicit STA sampling scheme using the overlap between the NBS and STA samples. It estimates density-ratio weights and draws simulated STA samples that are representative of the above-scale manufacturing population.
- **Main production-function result.** Using these simulated samples and the Gandhi–Navarro–Rivers (2020) gross-output production-function estimator, the paper constructs firm-level and aggregate revenue-based TFP measures that are comparable before and after 2007.
- **Main empirical result.** Aggregate manufacturing TFP growth falls sharply after 2007. The preferred baseline estimate is about **4.5% per year** in 1998–2007 versus only **1.1% per year** in 2007–2013.
- **Where the slowdown shows up.** The decline is broad-based across industries, ownership types, and regions. It is especially striking in the **private sector**, which continues to expand in size but experiences the weakest productivity growth after 2007.
- **Entry and dynamism.** A key part of the story is weaker market dynamism. Young firms become relatively less productive than incumbents, and the contribution of entrants to aggregate productivity growth falls sharply.
- **Why this paper matters.** It is both a data paper and a productivity paper. It gives a practical strategy for working around post-2007 Chinese firm-data problems and delivers a new micro-founded picture of China’s manufacturing slowdown.

2 Data and measurement (key objects)

2.1 NBS annual survey of above-scale industrial enterprises

- **Coverage before 2007.** From 1998 to 2006, the NBS survey covers all state-owned enterprises and all other industrial firms with revenue above 5 million RMB.
- **Coverage after 2007.** Starting in 2007, ownership is dropped as a sampling criterion and the survey covers firms with revenue above 5 million RMB regardless of ownership. In 2011, the threshold rises to 20 million RMB.

- **Ownership groups.** The paper aggregates detailed ownership codes into four groups: state-owned enterprises (SOEs), other domestic Chinese firms (non-SOEs), Hong Kong/Macau/Taiwan firms (HMT), and foreign-invested enterprises (FIEs).
- **Strength of the pre-2007 NBS data.** Earlier work shows that, up to around 2007, the NBS firm data aggregate closely to statistical yearbook totals and line up well with census data.
- **Post-2007 problems.** After 2007, the NBS data deteriorate sharply. Value added, intermediate inputs, and non-wage labor compensation are no longer reported; there is no 2010 micro sample; some provinces are missing in 2011; and key quantities such as output and employment appear increasingly overstated.
- **Economic interpretation.** Because the NBS data feed into China’s industrial national accounts, micro-level over-reporting in the NBS sample is relevant for interpreting broader macro concerns about overstated industrial growth.

2.2 STA annual tax survey

- **Purpose.** The STA survey is designed for tax administration rather than for statistical representativeness.
- **Always-surveyed firms.** Listed firms, large private corporations, firms affiliated with central or provincial governments, and various “focus firms” receiving special tax treatment are always included.
- **Sampled firms.** Other firms are selected by a stratified design based on 2-digit industry and firm size. The three size groups are defined by revenue cutoffs at 20 million RMB and 400 million RMB.
- **Key weakness.** Local tax offices can replace firms and adjust effective sampling intensity. So the final STA sample is **not representative**, and the actual sampling weights are unknown.
- **Key strength.** The STA data are less exposed to the local political incentives that seem to distort the NBS reporting after 2007.
- **Tax-reporting biases.** Because the data are collected for VAT administration, firms may overstate intermediate inputs to enlarge VAT deductions and may understate revenue to reduce taxes. The paper argues that tax reforms over time likely reduced these biases, so post-2007 TFP growth estimates from the STA side should be interpreted as an **upper bound**.

2.3 Matching the two sources and observed discrepancies

- **Overlap.** The authors match firms across the two surveys using legal IDs and firm names. In a typical year, about one-third to one-half of firms in one sample also appear in the other.
- **Coverage example.** In 2007 the NBS sample has 312,055 firms and the STA sample has 269,659 firms; in 2013 the NBS sample has 319,673 firms and the STA sample has 246,655 firms.
- **Above-scale overlap.** A large majority of above-scale STA firms can be matched to the NBS in later years; this fraction exceeds 80% in the last three years of the sample.

- **Output discrepancy.** For matched firms, the ratio

$$\frac{\text{STA reported value}}{\text{NBS reported value}}$$

falls strongly over time for output. At the 25th percentile, it falls from 0.54 in 2007 to 0.26 in 2013.

- **Employment discrepancy.** The same ratio for employment falls from 0.81 at the 25th percentile in 2007 to 0.36 in 2013.
- **More reliable variables.** Paid-in capital, firm age, and export value match much more closely across the two sources than output or employment.
- **Geography and ownership.** Over-reporting in the NBS is much worse in some provinces (for example, several provinces in the Northeast and Central China) and is more severe for private firms than for foreign-invested firms.

2.4 Additional data, variable construction, and applications

- **Business Registry and Inspection Data.** These sources are used to infer the evolution of the number of active firms by size category and to study firm entry patterns, especially after 2007.
- **Capital stock.** Real capital is constructed by perpetual inventory using fixed assets and investment deflators.
- **Deflators.** Output deflators are based on the producer price index at the 2-digit industry level; input deflators are built from input-output tables.
- **Main empirical outcomes.** The paper studies aggregate TFP growth, heterogeneity by ownership, industry, and region, and the changing role of young firms in aggregate productivity growth.

3 Models and empirical specifications (all equations + interpretation)

3.1 Target moment: aggregate productivity as a weighted expectation

The object of interest is aggregate productivity in a target population T . For a firm with output y , inputs $\mathbf{x}$, and attributes $\mathbf{A}$, productivity is the residual from a production function. Let the productivity-relevant object be written as

$$\omega^* = g(y, \mathbf{x}, \mathbf{A}).$$

Then the target moment is the size-weighted productivity average

$$m_t^T(y, \omega^*(\mathbf{A})) = \int_y \int_{\mathbf{x}} \int_{\mathbf{A}} g(y, \mathbf{x}, \mathbf{A}) f_t^T(y, \mathbf{x}, \mathbf{A}) d\mathbf{A} d\mathbf{x} dy. \quad (1)$$

- **Interpretation.** If we knew the representative target distribution $f_t^T(\cdot)$, we could just average firm-level productivity with the correct population weights.

- **Problem.** After 2007, the representative target distribution is not observed directly because the NBS becomes unreliable and the STA is unrepresentative.

3.2 Density-ratio reweighting

The key step is to rewrite the target moment using the source sample S and a density ratio:

$$m_t^T(y, \omega^*(\mathbf{A})) = \int_y \int_{\mathbf{x}} \int_{\mathbf{A}} g(y, \mathbf{x}, \mathbf{A}) r_t(y, \mathbf{x}, \mathbf{A}) f_t^S(y, \mathbf{x}, \mathbf{A}) d\mathbf{A} d\mathbf{x} dy, \quad (2)$$

where

$$r_t(y, \mathbf{x}, \mathbf{A}) = \frac{f_t^T(y, \mathbf{x}, \mathbf{A})}{f_t^S(y, \mathbf{x}, \mathbf{A})}. \quad (3)$$

- **Interpretation.** The weight r_t tells us how much each STA observation should count to reproduce the representative NBS-like population.
- **Economic meaning.** Firms that are underrepresented in the STA relative to the target population get larger weights; overrepresented firms get smaller weights.

3.3 Constant versus time-varying weighting

If the STA sampling rule were stable over time, one could estimate a single time-invariant density ratio $r_{2007}(y, \mathbf{x}, \mathbf{A})$ using the overlap year 2007 and apply it to later years.

If the STA sampling rule changes over time, the paper imposes the assumption

$$\text{Prob}_t(S = 1 \mid y, \mathbf{x}, \mathbf{A}) = \text{Prob}_t(S = 1 \mid k, \mathbf{z}, \mathbf{A}), \quad (4)$$

where k is real capital and $\mathbf{z}$ is a vector of reliably measured variables that do not enter the production function directly, including the wage bill, fixed assets at original purchase price, paid-in capital, export status, and export value.

Under this assumption, the density ratio can be rewritten as a function of $(k, \mathbf{z}, \mathbf{A})$, yielding

$$m_t^T(y, \omega^*(\mathbf{A})) = \int_y \int_{\mathbf{x}} \int_{\mathbf{A}} g(y, \mathbf{x}, \mathbf{A}) r_t(k, \mathbf{z}, \mathbf{A}) f_t^S(y, \mathbf{x}, \mathbf{A}) d\mathbf{A} d\mathbf{x} dy. \quad (5)$$

The same logic applies for subgroup-specific productivity moments conditional on attributes $\mathbf{A} = \mathbf{a}$.

- **Interpretation.** The time-varying approach lets the reweighting adapt year by year, which is important if local offices change the actual STA sampling scheme.
- **Why the proxy variables matter.** Later NBS waves no longer report all variables cleanly, so the paper uses a set of consistently measured proxies to trace the representative distribution over time.

3.4 Estimating the density ratio

The paper uses the Least Squares Importance Fitting method of Kanamori et al. (2009). If the true density ratio is $r^*(\mathbf{v}) = f_T^*(\mathbf{v})/f_S^*(\mathbf{v})$, estimate $r(\mathbf{v})$ by minimizing a weighted squared distance. In

practice, approximate the ratio by

$$r(\mathbf{v}) = \sum_{c=1}^C \alpha_c \phi_c(\mathbf{v}),$$

where $\{\phi_c\}_{c=1}^C$ are kernel basis functions. The empirical optimization problem becomes

$$\min_{\alpha \in \mathbb{R}^C} \left[\frac{1}{2} \alpha' \hat{H} \alpha - \hat{h}' \alpha + \lambda \mathbf{1}' \alpha \right],$$

with a regularized unconstrained approximation of the form

$$\min_{\beta \in \mathbb{R}^C} \left[\frac{1}{2} \beta' \hat{H} \beta - \hat{h}' \beta + \frac{\lambda}{2} \beta' \beta \right], \quad \tilde{\beta}(\lambda) = (\hat{H} + \lambda I)^{-1} \hat{h}.$$

- **Implementation detail.** The paper uses Gaussian kernels and 1000 kernel centers.
- **Sampling strata.** Ideally, the density ratio is estimated separately for firms in the three STA size groups: 5–20 million RMB, 20–400 million RMB, and above 400 million RMB.
- **Data complications.** After the NBS threshold rises in 2011 and some variables go missing in 2009–2010, the time-varying weighting scheme has to be adapted. For example, the 2008 weighting function is reused for 2009 and 2010.

3.5 Simulating representative samples

- **Firm counts by size.** The paper uses the 2008 NBS census year together with Business Registry and Inspection Data to infer how the number of firms evolves in each size class between 2008 and 2013.
- **Why this is needed.** The raw NBS counts after 2010 imply an implausibly large increase in the number of large firms; the paper therefore constructs alternative size-category counts.
- **Simulation procedure.** For each year and industry-size cell, STA firms are sorted into 10 bins by estimated weight, and simulated samples are drawn so that the mass of firms across bins reproduces the target composition.
- **Multiple simulations.** The paper draws 5 simulated samples and averages later results across them.

3.6 Production function estimation

Firm-level gross output technology is

$$y_{it} = f(k_{it}, l_{it}, m_{it}) + \omega_{it} + \varepsilon_{it}, \quad \omega_{it} = \rho \omega_{i,t-1} + \eta_{it}. \quad (7)$$

- **Gross output.** Intermediate inputs m_{it} enter directly, so this is a gross-output rather than value-added production function.
- **Estimator.** The paper uses Gandhi–Navarro–Rivers (2020), which identifies the derivative of the production function with respect to materials from the first-order condition for materials and then integrates back to recover the production function.

- **Why this estimator.** It allows a flexible nonparametric technology, avoids imposing constant returns to scale, and uses the material-input first-order condition to discipline the intermediate-input elasticity.
- **Separate periods.** The production function is estimated separately for 1998–2007 and 2007–2013 so that technology and substitution patterns can differ across the two periods.

3.7 Aggregation and productivity growth

Firm-level productivity is the residual from the production function, and aggregate industry productivity is computed as the output-share-weighted average

$$\hat{\Omega}_t = \sum_i s_{it} \hat{\omega}_{it}.$$

Aggregate manufacturing productivity growth is then the output-weighted average of industry-level productivity growth rates.

- **Interpretation.** This is revenue-based TFP growth, aggregated using firms' output shares.
- **Important nuance.** Because the STA still suffers from some tax-related reporting biases, the post-2007 TFP estimates are interpreted as likely upper bounds.

3.8 Decomposition into old firms and young firms

To study the role of dynamism and entry, the paper decomposes aggregate productivity growth from year 0 to year t as

$$\bar{\omega}_t - \bar{\omega}_0 = \sum_{i \in OLD} s_{it} (\omega_{it} - \bar{\omega}_0) + \sum_{e \in YNG} s_{et} (\omega_{et} - \bar{\omega}_0). \quad (8)$$

- **Interpretation of the first term.** This captures the contribution of old firms. It includes continuing firms, shifts in output shares across old firms, and the contribution of exit because exit changes the composition of surviving firms.
- **Interpretation of the second term.** This captures the contribution of young firms, which stand in for entrants when true population entry/exit cannot be observed in the STA sample.
- **Operational definition.** The paper uses firm age rather than first appearance in the sample to distinguish young firms from incumbents.

4 Identification and instruments (what makes the estimates credible)

4.1 Why the standard data approach fails

- **Why NBS alone fails after 2007.** The NBS is representative, but crucial post-2007 variables are missing and the remaining output and employment measures appear increasingly inflated.
- **Why STA alone fails.** The STA has richer and more reliable firm-level accounting information, but it is not representative and its effective weights are unknown.

- **Implication.** Neither sample by itself can deliver credible, comparable post-2007 productivity estimates.

4.2 What identifies the representative reweighting strategy

- **Overlap year.** The two surveys overlap in 2007, so the authors can compare the joint distribution of firm variables in a representative target sample (NBS) and an unrepresentative source sample (STA).
- **Core identifying idea.** If the difference between the two samples can be summarized by a density ratio, then reweighting the STA by that ratio reconstructs the representative population.
- **Time-varying extension.** When the true STA sampling rule may change over time, the paper identifies reweighting through variables that are consistently measured in both data sources and that predict selection into the STA sample.

4.3 Why the validation exercises are persuasive

- **Matched-firm comparisons.** Reporting discrepancies are directly documented for the same firms observed in both surveys, which makes the data problem transparent rather than hypothetical.
- **Distribution checks.** Kernel densities for output and paid-in capital show that the simulated samples are much closer to the NBS than the raw STA.
- **Growth-rate checks.** The simulated samples imply much more plausible growth rates for output, employment, and capital than the raw NBS post-2007 sample.
- **Robustness to weighting choice.** Constant-weight and time-varying-weight simulations lead to similar qualitative conclusions, so the main slowdown result is not driven by one particular weighting design.

4.4 Why the production-function stage is credible

- **Flexible technology.** The use of a nonparametric gross-output production function reduces the risk that results are driven by strong functional-form assumptions.
- **Input elasticity discipline.** The materials first-order condition helps discipline the material elasticity, a parameter often problematic in China applications.
- **Cross-period robustness.** Recomputing TFP growth with production-function parameters estimated in the other subperiod changes aggregate results only slightly.

4.5 Main threats and limitations

- **STA measurement is not perfect.** Because the STA is a tax survey, intermediate-input over-reporting and revenue under-reporting remain possible.
- **Upper-bound issue.** If tax-related misreporting diminished over time, then measured post-2007 TFP growth may still overstate true TFP growth.

Table 1
Coverage of the NBS and STA samples.

Year	(a) NBS survey				(b) STA survey				
	No. of firms	Above scale (%)	New in sample (%)	Matched w/ STA (%)	No. of firms	Above scale (%)	New in sample (%)	Matched w/NBS (%)	Above & Unmatched (%)
2007	312,055	98.1	18.2	35.5	269,659	41.2	–	41.2	11.6
2008	382,813	97.8	29.0	34.8	306,985	45.1	28.7	43.5	11.5
2009	361,720	98.3	8.8	36.0	302,515	50.4	29.8	43.3	16.7
2010	–	–	–	–	309,815	56.3	36.4	–	–
2011	279,242	98.6	–	45.5	279,666	44.5	25.6	45.5	9.5
2012	288,627	98.3	12.8	44.0	255,689	50.3	28.2	49.8	11.0
2013	319,673	98.7	18.3	41.4	246,655	51.0	20.7	53.7	9.5

Notes: Observations are matched by name and legal entity ID (组织机构代码、法人代码). The “above-scale” cutoff rises from 5 million to 20 million RMB in 2011.

- **Sample-rotation issue.** The STA is not a panel of the full population, which is why entry and exit cannot be measured in the same way as in earlier NBS-based work.
- **Interpretation issue.** The paper convincingly establishes a sharp slowdown, but it does not fully pin down which combination of state policy, external demand, competition, or knowledge-spillover channels caused it.

4.6 Relation to the literature

- It extends the pre-2007 China productivity literature that relied on the NBS annual survey, especially Brandt, Van Biesebroeck, and Zhang (2012).
- Methodologically, it combines survey reweighting ideas from Hellerstein and Imbens (1999) and Kanamori et al. (2009) with the gross-output production-function identification of Gandhi, Navarro, and Rivers (2020).
- Substantively, it speaks to the literature on Chinese misallocation, state versus private firms, and declining business dynamism.

5 Tables and figures

5.1 Table 1: Coverage of the NBS and STA samples

Read:

- The NBS and STA are both large, but they do very different things. The NBS is closer to a representative industrial survey; the STA is a tax-administration sample with different inclusion rules.
- The overlap between the two samples is substantial enough to compare the same firms across the two sources and estimate the weighting scheme.
- The importance of the STA rises after 2011 because the NBS threshold jumps to 20 million RMB and because post-2007 NBS quality deteriorates.
- The table is the paper’s first demonstration that a hybrid strategy is feasible: the two datasets are different, but sufficiently connected to be used jointly.

5.2 Table 2 and Figure 1: Reporting discrepancies across the two surveys

Read:

Table 2
Ratios of reported values in the matched NBS-STA sample.

Year	(a) Output					(b) Employment				
	p10	p25	p50	p75	p90	p10	p25	p50	p75	p90
2007	0.14	0.54	1.00	1.00	1.08	0.43	0.81	1.00	1.05	1.33
2008	0.12	0.47	0.97	1.00	1.05	0.40	0.77	1.00	1.04	1.28
2009	0.08	0.32	0.88	1.00	1.15	0.35	0.75	1.00	1.05	1.40
2011	0.09	0.30	0.88	1.00	1.04	0.19	0.36	0.62	0.88	1.29
2012	0.09	0.32	0.91	1.00	1.04	0.19	0.36	0.60	0.87	1.43
2013	0.07	0.26	0.86	1.00	1.04	0.18	0.36	0.67	1.13	1.77
	(c) Paid-in Capital					(d) Fixed Assets				
	p10	p25	p50	p75	p90	p10	p25	p50	p75	p90
2007	0.50	1.00	1.00	1.00	1.03	0.36	0.97	1.00	1.00	1.04
2008	0.46	1.00	1.00	1.00	1.00	0.24	0.84	1.00	1.00	1.01
2009	-	-	-	-	-	-	-	-	-	-
2011	0.50	1.00	1.00	1.00	1.12	0.18	0.68	1.00	1.00	1.04
2012	0.60	1.00	1.00	1.00	1.00	0.19	0.78	1.00	1.00	1.02
2013	0.50	1.00	1.00	1.00	1.02	0.16	0.69	1.00	1.00	1.02
	(e) Export value					(f) Number of Exporters				
	p10	p25	p50	p75	p90	NTB	STA	BOTH		
2007	0.28	0.78	1.00	1.02	1.21	34,226	34,709	27,680		
2008	0.16	0.65	1.00	1.01	1.25	38,878	34,423	24,873		
2009	0.19	0.69	1.00	1.01	1.21	36,043	39,614	27,923		
2011	0.26	0.87	1.00	1.01	1.20	38,412	45,073	33,056		
2012	0.30	0.88	1.00	1.01	1.18	42,032	50,012	37,781		
2013	0.27	0.87	1.00	1.01	1.19	40,550	49,173	36,126		
	(g) Firm age					(h) Wage				
	p10	p25	p50	p75	p90	p10	p25	p50	p75	p90
2007	0.62	1.00	1.00	1.00	1.38	0.25	0.63	1.00	1.02	1.30
2008	0.75	1.00	1.00	1.00	1.33	0.18	0.50	0.98	1.00	1.23
2009	0.75	1.00	1.00	1.00	1.56	-	-	-	-	-
2011	0.82	1.00	1.00	1.00	1.17	0.17	0.49	0.91	1.00	1.25
2012	0.86	1.00	1.00	1.00	1.17	0.19	0.52	0.91	1.00	1.21
2013	0.88	1.00	1.00	1.00	1.12	0.16	0.47	0.86	1.00	1.18

Notes: Reported statistics are the ratio of the values for the same variable in both samples: (STA value)/(NBS value). Information on Paid-in Capital and the value of Fixed Assets at original purchase price is not reported in the 2009 NBS survey. Panel (f) presents the count of exporters with reported export value above 10,000 RMB in each sample and the number of those firms reported as exporters in both samples.

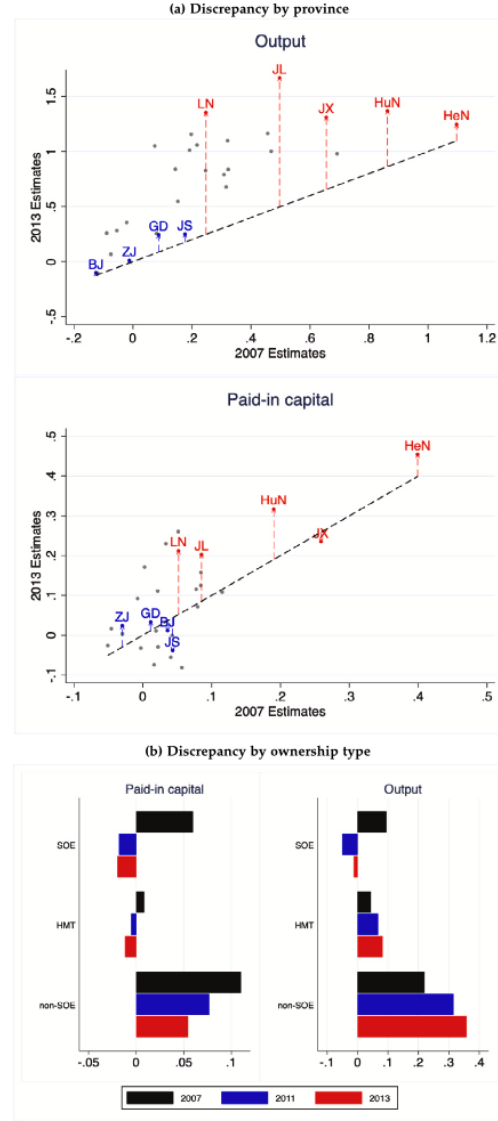


Fig. 1. Patterns in the reporting discrepancies.

(a) Notes: Discrepancy measures are obtained as coefficients on province dummies (Shanghai as reference) from annual regressions of the reporting discrepancy on firm characteristics that further include 2-digit industry-fixed effects and ownership type indicators. Each marker represents a province in mainland China. Arrows indicate the difference from the 45 degree line, i.e., the change in discrepancy from 2007 to 2013 relative to the change recorded for Shanghai. Provinces in blue (BJ-Beijing, ZJ-Zhejiang, GD-Guangdong, JS-Jiangsu) consistently show a low output discrepancy; provinces in red (LN-Liaoning, JL-Jilin, HuN-Hunan, JX-Jiangxi, HeN-Henan) either have very high discrepancy by 2013 or experienced a very large increase in discrepancy between 2007 and 2013.

(b) Notes: Discrepancy measures are obtained as coefficients on ownership type dummies (foreign-invested firms as reference) from annual regressions of the reporting discrepancy on firm characteristics that further include 2-digit industry-fixed effects and province indicators.

- The discrepancy problem is real and worsening. For matched firms, the output and employment ratios drift far below one over time, indicating rising overstatement in the NBS relative to the STA.
- The worsening is especially strong in the lower tail of the matched-firm distribution, meaning that for a nontrivial set of firms the gap becomes very large by 2013.
- The distortions are not random. Figure 1 shows they are systematically related to province and ownership, with some provinces showing especially severe deterioration and private firms exhibiting the largest output discrepancy.
- This is the core factual justification for abandoning the post-2007 NBS data as the sole basis for productivity analysis.

5.3 Figure 2 and Figure 3: Recovering the STA sampling scheme

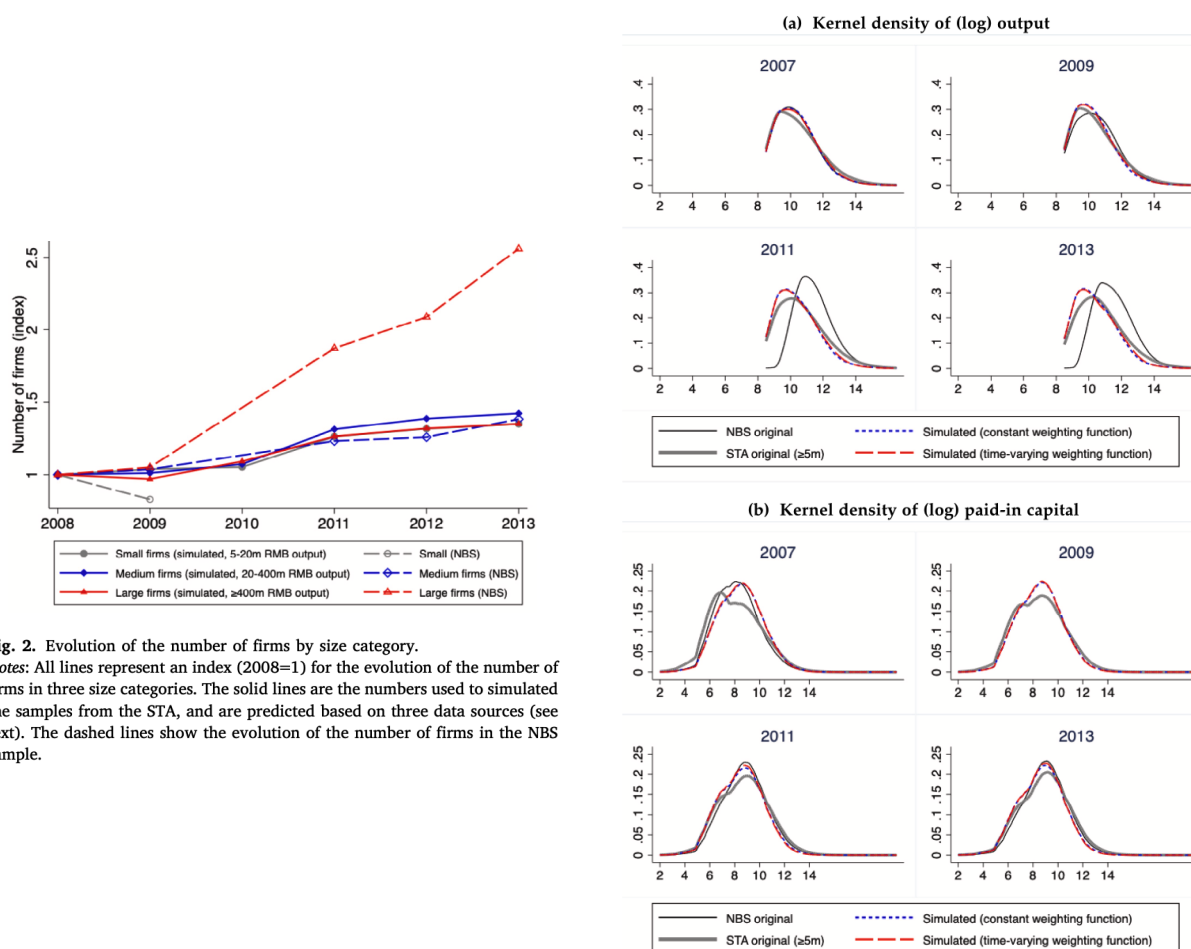


Fig. 3. Marginal distributions of selected variables in the different samples.

Read:

- Figure 2 shows that the raw NBS size distribution after 2008 implies an implausibly sharp rise in very large firms. The alternative reconstructed counts are much more stable and believable.

- Figure 3 shows that the simulated STA samples line up much more closely with the NBS target distributions than the raw STA does.
- The time-varying weighting scheme is particularly useful once the NBS threshold changes and once some NBS variables disappear.
- These figures validate the paper’s central data-construction step: the simulated STA sample is designed to be both *cleaner* and *representative*.

5.4 Table 3 and Figure 4: How the simulated samples change the picture

Table 3

Annualized growth rates of output and input variables (2007–2013).

	Value added (nominal)	Gross output (nominal)	Employment (persons)	Fixed asset (purchasing value)	Real capital (constructed)
NBS above-scale survey	–	15.4	11.3	16.2	13.7
STA unweighted	10.4	13.1	1.1	12.2	10.3
Simulated (constant w.f.)	10.4	12.5	5.1	14.6	12.8
Simulated (time-varying w.f.)	9.6	11.5	4.0	13.4	11.5
NBS Yearbook (above-scale)	12.0	15.9	3.6	15.4	

Notes: For the firm-level samples, we first aggregate variables for all manufacturing firms by year and then calculate a single annualized growth rate over the full period. Value added and gross output are in current prices, and real capital values are constructed using a perpetual inventory method (see Brandt et al., 2014). For the NBS Yearbook, we report the geometric mean of the reported annual growth statistics. They are for the entire industrial sector, including mining and utilities. The capital statistic is the growth in reported fixed asset at the original purchasing value. The NBS Yearbook reports a growth in industrial GDP for all firms, not limited to above-scale firms, of 11.5%.

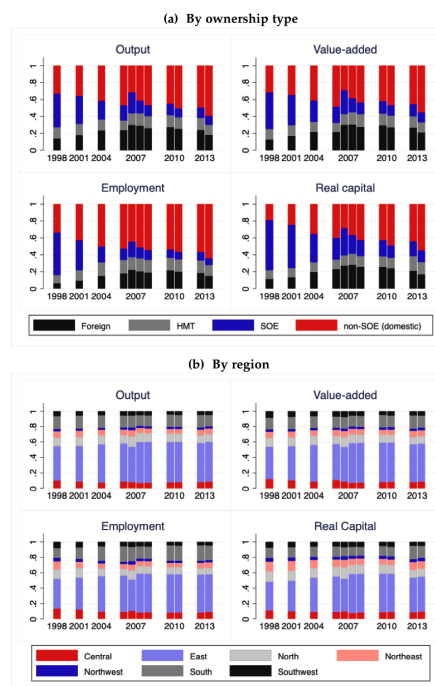


Fig. 4. Evolution of the sample composition.

Notes: The single bars in 1998, 2001, and 2004 are for the NBS sample. The four bars in 2007, from left to right, correspond to the NBS sample, the unadjusted STA sample, the simulated sample with the constant weighting function, and the simulated samples with the time-varying weighting scheme. The two bars in 2010 and 2013 are based on the two simulated samples, constant weighting function on the left and time-varying weighting function on the right.

Read:

- The post-2007 NBS sample implies implausibly fast growth in gross output, employment, and fixed assets. For example, nominal gross output grows 15.4% per year in the NBS but only 13.1% in raw STA and 11.5% in the preferred time-varying simulated sample.
- Employment growth is especially revealing: 11.3% per year in the NBS versus only 1.1% in raw STA and 4.0% in the preferred simulated sample.
- Figure 4 shows that ownership composition changes substantially over time, especially the rise of domestic private firms and the continued decline of the state share.
- Regional composition changes little, so the big post-2007 story is much more about ownership and dynamism than about dramatic regional reallocation.

5.5 Figure 5: Production elasticities before and after 2007

Read:

- Material elasticities rise in every 2-digit industry after 2007.
- Capital elasticities change little, while labor elasticities fall in most industries.
- Returns to scale are slightly higher in the later period and are close to one in almost every industry after 2007.
- The figure says that measured post-2007 production technology looks more materials-intensive and slightly more constant-returns-like than before.

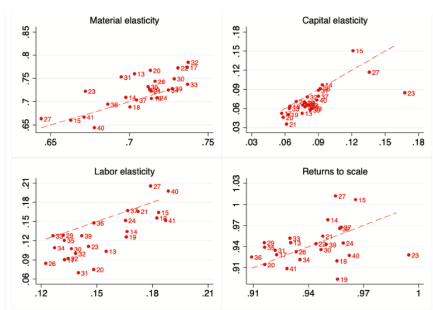


Fig. 5. Output elasticities estimated on 1998–2007 and 2007–2013.
Notes: The first three panels show the output elasticities for the three inputs estimated using a non-parametric production function. The horizontal axis shows the estimates for 1998–2007 on the NBS data and the vertical axis the estimates for 2007–2013 on the samples simulated with a time-varying weighting function. The fourth panel shows returns to scale calculated as the sum of the three elasticities. All values are the median across all firms in a 2-digit CIC industry (codes indicated next to the markers). The dashed line is the 45-degree line. Comparable results for samples based on a constant weighting function are in Fig. B.1 in Appendix.

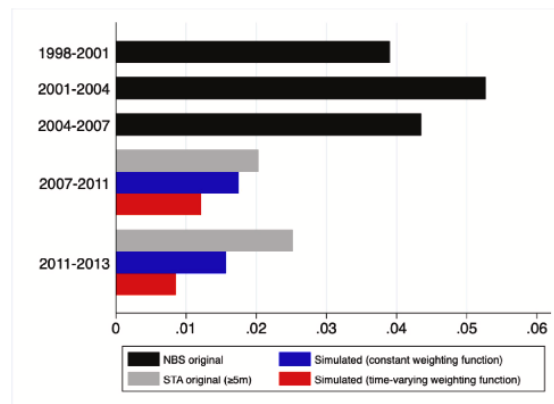


Fig. 6. Annualized aggregate productivity growth in China's manufacturing.

5.6 Figure 6: Aggregate TFP growth collapses after 2007

Read:

- This is the paper's headline figure. Aggregate manufacturing TFP growth is about 4.5% per year in 1998–2007 but only 1.1% per year in 2007–2013 in the preferred baseline.
- Using the raw STA data without reweighting yields higher post-2007 growth than the simulated representative samples, which shows why the weighting step matters.
- Constant-weight and time-varying simulated samples give similar slowdown patterns, though the preferred time-varying series is lower.
- The slowdown is therefore not a fragile artifact of a single sample construction choice.

5.7 Figure 7 and Figure 8: Ownership and industry heterogeneity

Read:

- By ownership, all groups slow down after 2007, but the slowdown is especially pronounced for private domestic firms, whose post-2007 productivity growth falls to only a bit above 1% per year.

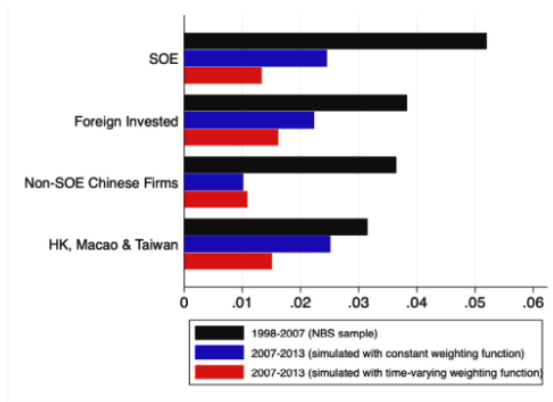


Fig. 7. Productivity growth by ownership type.

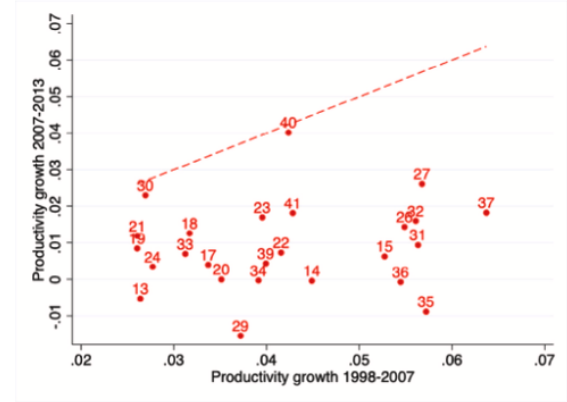


Fig. 8. Productivity growth by industry.

Notes: The results for 2007–2013 use samples simulated with the time-varying weighting function. Results based on the constant weighting function are in Fig. B.3 in Appendix. The exact productivity growth estimates by industry are reported in Table B.1 in Appendix. The dashed line is the 45-degree line.

- SOEs had strong pre-2007 productivity growth partly because restructuring removed the lower tail of inefficient state firms. After 2007, their productivity growth also drops sharply.
- By industry, the slowdown is nearly universal. Average industry productivity growth falls from 4.4% to 1.4%.
- Communications equipment and electronics (CIC 40) is the big outlier, still posting very strong growth after 2007, while several machinery and metal-product industries have near-zero or negative growth.

5.8 Figure 9, Figure 10, and Figure 11: Regional patterns and convergence

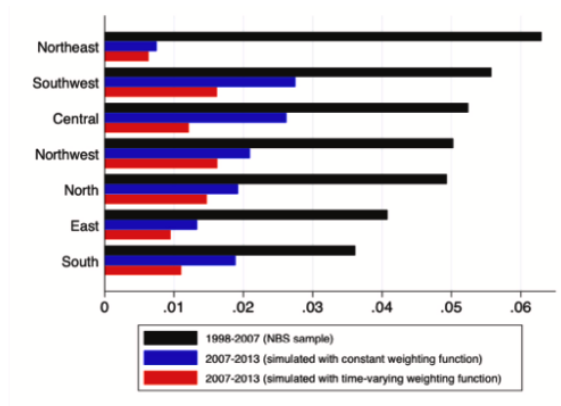


Fig. 9. Productivity growth by region.

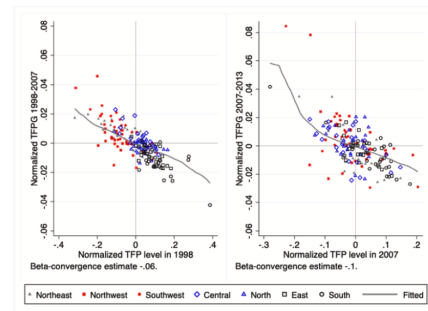


Fig. 10. Persistent productivity convergence across regions.
Notes: Each point represents an industry by region combination. TFP levels and growth rates are both demeaned across all regions within each industry. The graph in the right panel for 2007–2013 is based on simulated samples with the time-varying weighting function.

Read:

- Before 2007, TFP growth is strongest in the Northeast, Southwest, and Central regions and weakest in the South and East, though still robust everywhere.

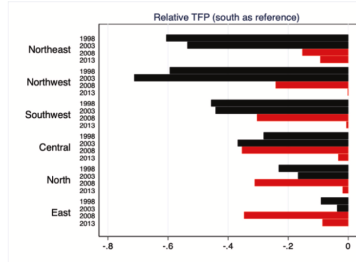


Fig. 11. Evolution of productivity levels across regions.
Notes: The relative TFP level of each region is computed as a weighted average of productivity differences from the South (the reference region) across industries. The weights are two-digit SIC industry gross output shares at the national level. Estimates for 1998 and 2003 are based on the NBS above-scale sample, while estimates for 2008 and 2013 utilize simulated samples with the time-varying weighting function.

- After 2007, TFP growth weakens sharply in all regions, especially in the Northeast; the South and East grow at only about 1% per year.
- Figure 10 shows continued productivity convergence across regions within industries, with beta-convergence around 6% before 2007 and 10% after 2007.
- But Figure 11 shows why convergence is not enough to rescue aggregate growth: the remaining regional gap relative to the South is already quite small by 2013, leaving limited room for future catch-up to generate large aggregate gains.

5.9 Figure 12, Figure 13, Table 4, and Table 5: The weakening role of entrants

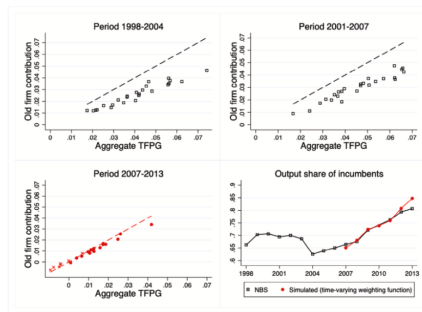


Fig. 12. Contribution to aggregate productivity growth of old firms (> 6 years).
Notes: The results for 2007-2013 use samples simulated with the time-varying weighting function. X markers represent 2-digit industries where young firms contribute negatively to industry-level productivity growth. From left to right, these are C129 (rubber products), C135 (general machinery), C133 (food processing), C136 (special equipment), C134 (metal products), C114 (food manufacturing) and C139 (electrical machinery).

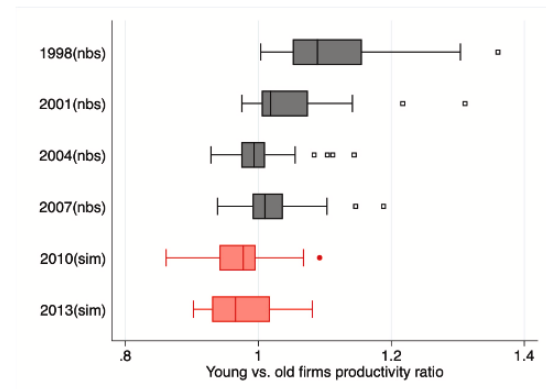


Fig. 13. Declining relative productivity of young firms (≤ 6 years old).
Notes: The box plot summarizes the distribution of relative productivity of young firms versus incumbents across 2-digit industries. The shaded box represents the inter-quartile range, and the vertical line in the box the median. The simulated samples are based on the time-varying weighting function. The change after 2010 is sharper if we define young firms as no more than 3 years old, as shown Fig. B.4 in Appendix.

Read:

- Figure 12 shows that young firms contribute much less to aggregate productivity growth after 2007. The share of aggregate TFP growth accounted for by old firms rises from about 69% to 85%, so the contribution of young firms falls from roughly 31% to 15%.
- Figure 13 shows that young firms become much less productive relative to incumbents. In early years, new entrants are often more productive than incumbents; by 2010 and 2013 they are usually less productive.

- Table 4 shows the entry rate among above-20-million-RMB NBS firms falling from 8.0% in 2007 to 4.8% in 2013. The composition of entrants also changes dramatically toward domestic private firms.
- Table 5 shows a collapse in foreign-invested firm entry: average annual new FIE entries fall from 19,852 in 2000–2007 to 6,062 in 2008–2014 and then to 3,419 in 2015–2018.

Table 4
Falling entry rates in the NBS survey.

Year	Total	Entry rate(%)	Share of new entrants (%)			
			non-SOE	SOE	HMT	Foreign
1998	48,815	7.4	52.9	17.0	14.0	16.2
1999	50,486	6.7	57.5	16.8	13.0	12.8
2000	54,613	5.8	61.6	13.1	12.6	12.8
2001	59,261	7.8	67.0	10.9	11.6	10.5
2002	67,256	7.1	69.0	8.1	11.9	11.1
2003	81,137	7.6	69.0	6.3	12.3	12.4
2004	107,327	11.9	69.1	4.3	12.1	14.5
2005	125,391	8.9	72.3	4.4	10.5	12.9
2006	150,006	8.2	73.0	3.6	10.0	13.3
2007	183,341	8.0	76.3	3.0	9.3	11.4
2008	215,976	8.1	81.1	3.5	7.0	8.5
2009	224,041	5.6	86.7	3.3	5.0	5.0
2011	275,365	5.8	90.8	2.7	3.4	3.1
2012	283,841	5.2	89.7	2.5	4.3	3.6
2013	315,762	4.8	91.7	1.9	3.6	2.8

Notes: Number of firms with reported revenue above 20 million RMB in the NBS annual firm survey. Entrants are firms new to the sample that were established at most one year earlier. The entry rate is the number of entrants divided by the number of active firms at the beginning of the year multiplied by 100.

Table 5
Falling entry rates for foreign-invested firms in manufacturing.

Period	Total	Light	Heavy	Advanced
1992–1999	21,790	11,121	7164	3506
2000–2007	19,852	8631	6914	4307
2008–2014	6062	2561	1480	2021
2015–2018	3419	1537	765	1117

Notes: Average number of new entrants per year as defined by their year of establishment, irrespective of size. Light, heavy and advanced are defined at the CIC 2-digit level, and described in the Appendix.

Source: Business Registry of China.

5.10 Table 6 and robustness checks

Read:

- The aggregate post-2007 slowdown is robust to using production-function parameters estimated on the pre-2007 sample and vice versa.
- For example, post-2007 aggregate TFP growth is still only 1.5% when evaluated using the pre-2007 production function parameters, versus 1.1% in the baseline.
- The broad message is that the slowdown is not being mechanically produced by changes in estimated elasticities alone.
- Appendix checks on value-added misreporting reinforce the paper’s interpretation that, if anything, the post-2007 slowdown may be understated rather than overstated.

Table 6
TFP growth based on alternative production function estimates (%).

Production function parameters:	Period and firm sample used		
	NBS 1998–2007	STA 2007–2013	
		Constant w.fct.	Time-varying w.fct.
NBS 1998–2007	4.5	1.5	1.5
STA 2007–2013			
- Constant weighting fct.	4.2	1.7	
- Time-varying weighting fct.	4.2		1.1

Notes: The statistics on the diagonal are the baseline estimates where TFP growth for each period is calculated using the production function parameters estimated on the same period. The off-diagonal statistics use production function parameters from a different period than the period over which TFP growth is calculated.

6 Short summary

- **Conclusion.** China’s manufacturing productivity growth slowed dramatically after 2007, and the slowdown is visible across sectors, ownership groups, and regions.
- **Identification insight.** Reliable post-2007 inference is possible only by combining the NBS and STA data and reweighting the STA to match the representative population.
- **Economic lesson.** The slowdown is not mainly about the state directly taking over manufacturing. It is more about a broad loss of dynamism, especially among private firms and new entrants.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that the standard post-2007 NBS firm data are too distorted to be used directly for productivity analysis.
- It shows how to use the STA tax survey, despite its nonrepresentative design, to recover representative firm samples and estimate comparable production functions and TFP over time.
- Once this correction is made, China’s manufacturing slowdown is much sharper than one would infer from the raw NBS data alone.
- The slowdown is broad-based, but the private sector and the entry margin stand out as the key places where dynamism weakens most.

7.2 Economic intuition

- In the early 2000s, China’s manufacturing sector still had many sources of dynamism: restructuring of inefficient state firms, strong export demand, and productive entry by younger and often private firms.
- After 2007, these forces weaken. Even though private firms continue to absorb resources and expand their output share, their productivity growth slows markedly.
- At the same time, young firms stop being engines of productivity growth. Fewer enter, and those that do enter have lower relative productivity than previous cohorts.
- The result is a manufacturing sector that still reallocates and still converges regionally, but no longer generates rapid aggregate productivity growth.

7.3 Takeaway

This paper is best understood as a bridge between data repair and substantive productivity economics. Its technical contribution is to make post-2007 Chinese manufacturing data usable again by combining representative but distorted NBS data with cleaner but nonrepresentative STA data. Its economic contribution is to show that the post-2007 period is not merely one of slower growth in a few sectors;

it is a broad decline in manufacturing dynamism. The most important qualitative message is that the private sector kept growing in size while losing productivity momentum, and that the contribution of young entrants to aggregate productivity growth deteriorated sharply. Any explanation of China's post-2007 growth slowdown has to confront these facts.